LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to Welding and types of welding technologies
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to Welding and types of welding technologies
5. Specific Objectives	31.1.3.1 Trainee will be able to perform various repair works on aircraft structure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of welding symbols and joint types; Safety briefing on PPE; Instructor supervision of theoretical concepts
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of welding in aircraft maintenance?
(b) Name one common material used in aircraft structural repair.
(c) Why is understanding welding types essential for an air engineer?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic principle of welding • Fundamental requirements for a sound weld • Role of welding in aircraft structural repair • Importance of filler metals and base metals • Basic classification of welding processes	L & Ds	PPT, IB	15 min
(b)	• Overview of Fusion Welding technologies • Arc welding methods: SMAW, GMAW, GTAW • Solid State and Resistance welding types • Comparison of manual vs automated welding • Selection criteria for aircraft repair welding	L & Ds	PPT, PDF	20 min
(c)	• Introduction to welding symbols and notation • Standard joint configurations in aircraft structures • Basic defects and their visual identification • Preparation requirements for different joint types • Post-weld inspection basics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between fusion and solid-state welding in one sentence.
(b) Identify two critical factors when selecting a welding process for aluminum aircraft structures.
(c) Describe the function of a welding symbol in a maintenance drawing.

12. Home Assignment and Follow Up	Prepare a brief note on the specific welding requirements for repairing a damaged aluminum skin panel, referencing the procedures to be covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Safety precautions to be observed during Gas/ Arc welding, Gas cutting and hot working	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min